

Anthony Meza

Ph.D. Candidate

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EDUCATION

Massachusetts Institute of Technology

Ph.D. in Physical Oceanography

Cambridge, MA

2021 - 2026 (Expected)

University of California, Irvine

B.S. in Mathematics - Concentration in Data Science

Irvine, CA

2018-2021

EXPERIENCE

Woods Hole Oceanographic Institution

Graduate Research Assistant

Sep 2021–Present

Woods Hole, MA

- Used numerical ocean and climate models to investigate mechanisms controlling ocean circulation and heat transport
- Analyzed satellite products using statistical methods (i.e., multivariate, time series, and spatial analysis), to identify links between coastal sea surface temperatures and extreme California precipitation events
- Developed Python and Julia tools to process, visualize, and interpret terabytes of climate data in high-performance computing environments

Foundation for Resilient Societies

Technical Consultant

Jan 2025-Feb 2025

Cambridge, MA

- Ran and debugged Strategic Energy & Risk Valuation Model (SERVM) simulations to assess U.S. electrical grid capacity adequacy under varying generation scenarios (e.g., solar adoption).
- Led a team of 12 undergraduate electric grid modeling interns for 4 weeks to develop an internal user guide for running SERVM experiments and interpreting model output.

Los Alamos National Laboratory

Parallel Computing Summer Fellow

Jun 2021–Aug 2021

Los Alamos, NM

- Implemented and evaluated reduced-precision in the Energy Exascale Earth System Model (E3SM), written in Fortran, to reduce computational cost and energy consumption in global climate simulations

PROJECTS

Atmospheric Physics using AI

Kaggle Competition Participant

Apr 2024–Jul 2024

Remote

- Ranked 230/694 on the public leaderboard for emulating subgrid-scale atmospheric processes
- Developed machine learning architectures with physical constraints in PyTorch and TensorFlow
- Optimized data pipelines for GPU-accelerated training, significantly increasing model throughput

xbuoy

Personal Project

- Developed *xbuoy*, a Python package for querying National Data Buoy Center observations and converting irregularly sampled buoy data to NetCDF and Zarr

SKILLS

Languages: *Programming:* Python, Julia, MATLAB; *Human:* English, Spanish

Scientific Computing: Scientific Python (xarray, PyTorch, Tensorflow) and Julia (Optimization.jl, JuMP.jl)

HPC & Dev Tools: Unix, HPC job schedulers (e.g., Slurm), Dask, Git, GitHub